Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | *Durable non-precious metal electrodes for large-area AEL/AEMEL operation under high-pressure and high-temperature conditions* | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Energy & Resources* | | *Mechanical・Materials* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *New & Renewable Energy* | *Hydrogen* | *Metal Materials* | *Materials Processing Technologies* |
| Concept of Proposed Technology | The proposed technology targets non-precious metal–based electrodes (Ni, Ni-Fe) for alkaline (AEL) and anion exchange membrane electrolysis (AEMEL), combining scalable electrode architectures (large-area up to 2,500 cm²) with systematic electrochemical validation for industrial operation. The electrodes are designed and assessed for stable performance under elevated pressure (up to 18 bar), high-temperature alkaline conditions (up to 80 °C), and dynamic load profiles (20–100%), enabling efficient operation under fluctuating renewable power input. Fraunhofer IFAM contributes advanced electrochemical characterization and accelerated stress testing (AST), leveraging test infrastructure capable of extended pressure/temperature conditions (up to 80 bar / 120 °C) for accelerated lifetime and failure-mode assessment. | | | |

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| **Rationale for Joint R&D Project** | Cost-effective and durable electrode technologies are essential for the large-scale deployment of alkaline (AEL) and anion exchange membrane electrolysis (AEMEL). Although non-precious metal–based electrodes such as Ni and Ni-Fe show high potential to replace precious-metal solutions, their performance stability under elevated pressure, high temperature, and dynamic operating conditions is not yet sufficiently validated for industrial application.<br><br>This joint R&D project combines industrial electrode development and manufacturing expertise from the Korean SME with advanced electrochemical testing, durability evaluation, and materials optimization capabilities at Fraunhofer IFAM. Such comprehensive validation—particularly under high-pressure and high-temperature alkaline conditions—cannot be achieved by a single partner alone. The collaboration enables systematic performance assessment, accelerated stress testing, and targeted improvement of large-area electrode concepts.<br><br>By integrating complementary expertise and validating the technology against international requirements, the project reduces technical risk, accelerates technology maturation, and supports the development of scalable, market-relevant electrode solutions for global hydrogen markets. |
| **Key Words** | 1. Non-precious metal electrodes<br>2. Alkaline electrolysis<br>3. AEM electrolysis<br>4. Accelerated stress testing (AST)<br>5. Large-area electrode scale-up |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 4 ) | End of the Project | Level ( 6 ) |
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| R&D Objectives | The objective of this R&D project is to develop and validate non-precious metal–based electrodes (Ni, Ni-Fe) for alkaline (AEL) and anion exchange membrane electrolysis (AEMEL) that combine high electrochemical performance with industrial durability and scalability. The project aims to optimize electrode composition, microstructure, and coating or fabrication routes to enable stable operation under elevated pressure, high temperature, and dynamic load conditions. A further objective is to demonstrate large-area electrode concepts suitable for direct integration into electrolyzer cells and stacks, supported by systematic electrochemical characterization and accelerated stress testing, targeting maturation from TRL 4 to TRL 6 (pilot-scale validation). | | | |
| R&D Target Specifications | The project targets electrode operation at up to 18 bar pressure and 80 °C in alkaline media, with stable performance over a dynamic load range of 20–100% in both AEL and AEMEL-relevant single-cell configurations. For AEMEL validation, selected commercial reference membranes/ionomers will be used to clearly attribute performance and degradation effects to the electrode technology. Target electrochemical performance is a single-cell voltage ≤ 2.0 V at 0.5 A $cm^{-2}$ (80 °C, 1 bar) and ≤ 2.15 V at 1.0 A $cm^{-2}$ (80 °C, up to 18 bar), with Faradaic efficiency ≥ 98%. Durability targets include a voltage degradation rate ≤ 20 μV $h^{-1}$ (≤ 40 mV per 2,000 h equivalent) under defined long-term and AST protocols. Large-area electrodes up to 2,500 $cm^2$ will be produced with ≤ ±5% performance variation across the active area and demonstrated in a short-stack / pilot-scale test to validate TRL 6 readiness. | | | |
| R&D Plan | The project will be implemented through two program phases in accordance with the TIPA call: Phase 1 (Pre-study, 6 months) and Phase 2 (Main R&D, 30 months). Work packages (WPs) are designed to connect manufacturable electrode concepts with rigorous electrochemical validation, durability assessment, and scale-up toward pilot-scale demonstration.<br><br>Phase 1 – Pre-study (Months 1–6)<br>WP1 (M1–3): Electrode concept definition and feasibility assessment (Ni / Ni-Fe electrodes for AEL and AEMEL; selection of scalable coating/fabrication routes; selection of reference cell components for fair benchmarking, incl. commercial AEM membranes/ionomers; fabrication of first samples; baseline half-cell and single-cell testing).<br>Milestone M1: Selection of electrode concepts for durability screening.<br><br>WP2 (M4–6): Benchmarking and early durability screening (initial AST screening; identification of key degradation mechanisms; definition of target test matrix including pressure/temperature and dynamic load profiles; refinement of Phase 2 targets and risk mitigation).<br>Milestone M2: Feasibility report and detailed Phase 2 roadmap (incl. data package for Phase 2 application). | | | |

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| | Phase 2 – Main R&D (Months 7–36)<br>WP3 (M7–24): Performance optimization and accelerated stress testing (iterative improvement of composition/structure; electrochemical validation under up to 18 bar and 80 °C; dynamic operation 20–100%; post-mortem analysis; optional higher-stress protocols using IFAM infrastructure up to 80 bar / 120 °C on selected samples to accelerate lifetime assessment).<br>Milestone M3: Demonstration of stable performance with low degradation under defined long-term/AST protocols.<br><br>WP4 (M25–32): Large-area electrode scale-up and validation (manufacturing of electrodes up to 2,500 cm²; assessment of performance uniformity and reproducibility; integration into single-cell or short-stack hardware; iteration with manufacturability feedback).<br>Milestone M4: Validated large-area electrode prototypes ready for pilot-scale testing.<br><br>WP5 (M33–36): Pilot-scale demonstration and TRL 6 validation (short-stack / pilot-scale testing using large-area electrodes; validation against target specifications under representative operating conditions; final documentation and exploitation roadmap).<br>Milestone M5: TRL 6 demonstration – pilot-scale prototype production and performance evaluation.<br><br>Overall outcome: Non-precious metal electrode prototypes for AEL/AEMEL with validated durability and scalability, ready for subsequent system-level integration and commercialization. |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | Fraunhofer IFAM will be responsible for the applied research, electrochemical evaluation, and durability validation of the proposed non-precious metal electrode technologies. This includes the design of test protocols, execution of advanced electrochemical characterization and accelerated stress testing under high-pressure and high-temperature alkaline conditions, and analysis of performance degradation mechanisms. Fraunhofer IFAM will provide access to its state-of-the-art electrochemical laboratories, high-pressure test infrastructure, and materials characterization facilities, and will lead the optimization of electrode compositions and structures based on experimental results. In addition, IFAM will coordinate the technical R&D roadmap, contribute to milestone evaluation, and support knowledge transfer toward scalable, market-oriented solutions. |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will contribute its industrial expertise in electrode design, manufacturing, and scale-up, focusing on the development of Ni- and Ni-Fe-based electrode materials suitable for AEL and AEMEL systems. The SME will be responsible for fabricating electrode samples and large-area prototypes (up to 2,500 $cm^2$) using scalable production routes and ensuring compatibility with electrolyzer system requirements. The SME will actively participate in performance evaluation and validation activities by providing manufacturing feedback, adapting designs based on test results, and supporting prototype integration. Furthermore, the SME will commit personnel, materials, and production resources to the project and collaborate closely with Fraunhofer IFAM to prepare the technology for subsequent commercialization and market entry. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The joint R&D project is expected to deliver validated, non-precious metal electrode technologies (Ni, Ni-Fe) for alkaline and anion exchange membrane electrolysis, demonstrated at TRL 5 under application-relevant operating conditions. Key technological outcomes include stable electrode performance at elevated pressure and temperature, low degradation rates under dynamic load operation, and successful demonstration of large-area electrode prototypes suitable for electrolyzer cell and stack integration. These results will significantly reduce technical risk associated with industrial deployment of non-PGM electrodes.<br><br>From a commercialization perspective, the project will enable a cost-effective alternative to precious-metal-based electrodes, supporting reduced material costs and improved supply-chain resilience. The validation of scalable manufacturing routes and large-area formats will accelerate technology transfer toward pilot and commercial electrolyzer systems, strengthening the market readiness of the Korean SME partner and facilitating entry into European and global hydrogen markets.<br><br>At a broader level, the project contributes to the competitiveness of the hydrogen industry by supporting lower-cost green hydrogen production and improved electrolyzer reliability. The collaboration strengthens international R&D linkages between Korea and Europe, promotes industrial innovation among SMEs, and supports national and global strategies for decarbonization and sustainable energy systems. |

## 6. References

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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